RCD’s principle & basic application

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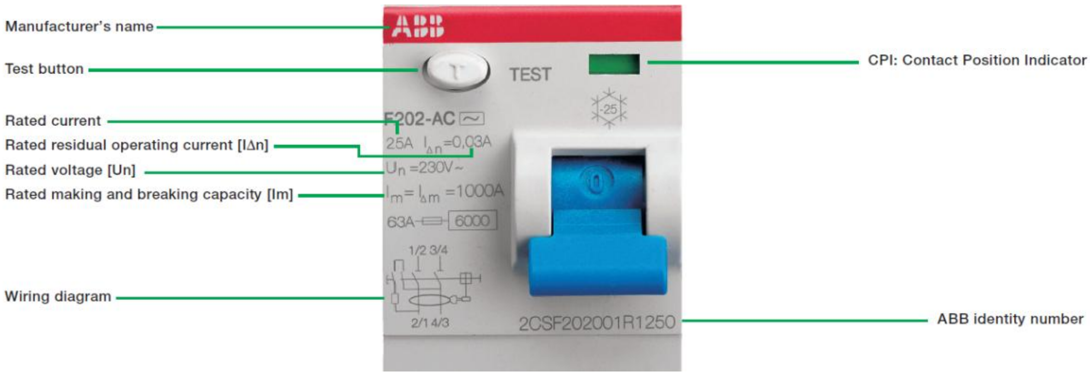
What's the RCD?

It is not clear when and by whom the first RCD was developed, but it certainly appeared on the market in the 1950s. It was initially used by some utility companies to fight "energy theft" who used to drag some energy out of the line.

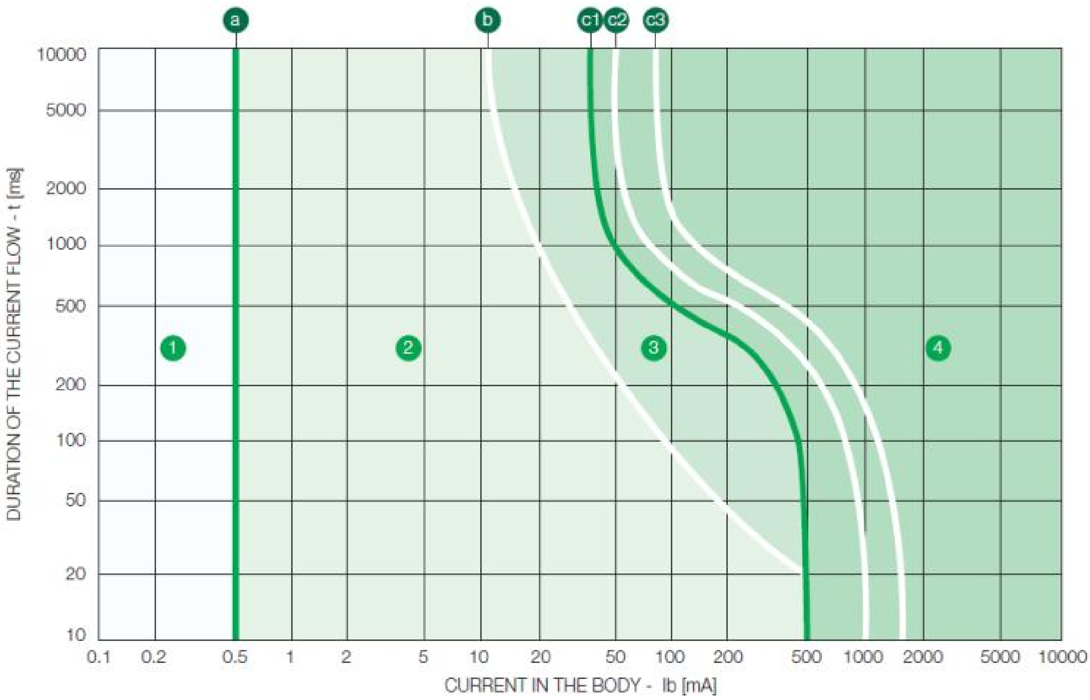
What's it looks like? Why we use it?

Nowadays RCD family was used widely in LV to protect against electrocution and fire hazard. PV industry also need to has this function as IEC, EN and UL required.

Please find one normal RCD(courtesy of ABB) below.



The IEC 60479-1 Standard inllustrated the"effects of alternating current(15~100Hz) on the human body". There're 4 zones to define the effects, which only Zone1 and Zone 2 are "usually safe" for human beeing.



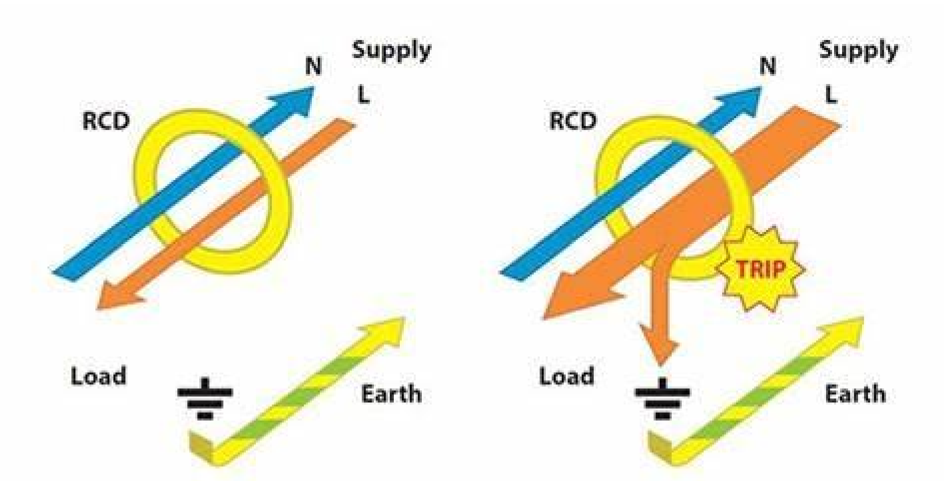
Effects of Zone 1&2 above as below,

Zone	Effects
1	Usually no reaction
2	Usually no harmful physiological effects

How’s the RCD working?

RCDs use the principles of Kirchhoff’s first law:

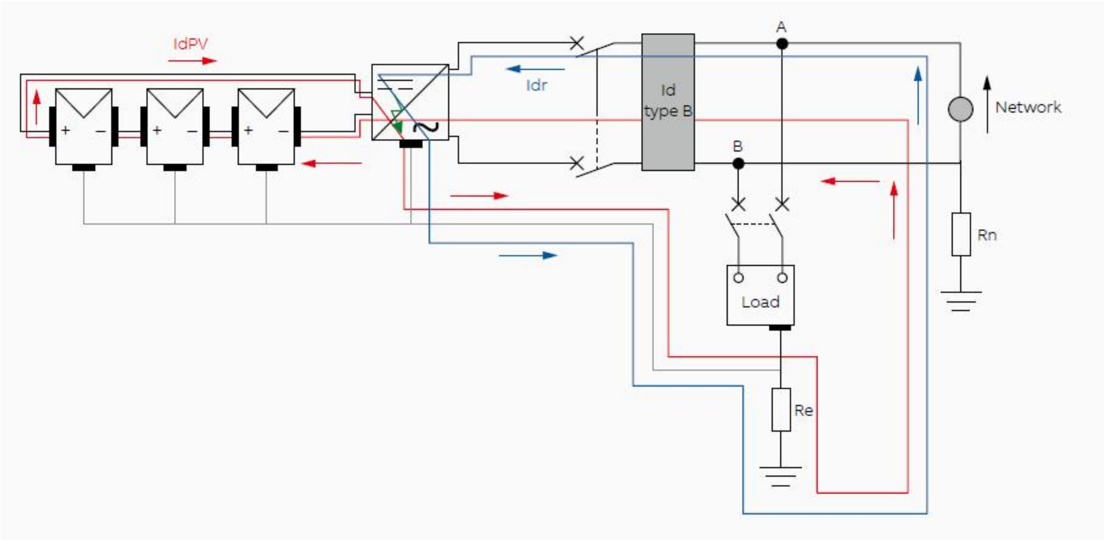
That is, the algebraic sum of the currents flowing to and from a node in a circuit should be equal to zero. In the event of an insulation fault to earth, a fault current (residual current) flows back to the source of the current via the earth and not via the current carrying conductors. This induce a current flow in the RCD trip circuit, i.e. the sum of the currents flowing through the RCD no longer equal zero.



If the effective value of the residual current $I_{\Delta n}$ exceed 50% of the designed operating level of the RCD (Rated residual current $I_{\Delta n}$), the RCD will trip and disconnect the load from supply.

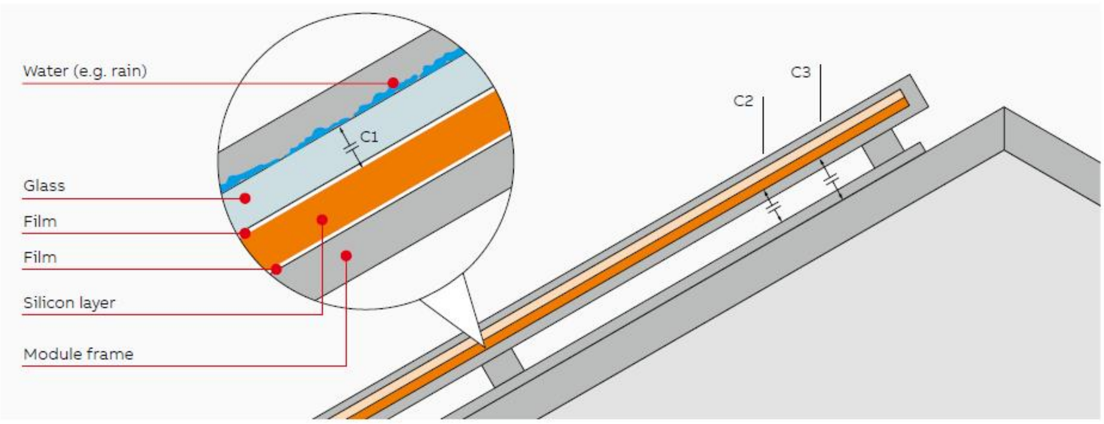
Where do the leakage current come from?

Apart from the fault current due to indirect contact in AC&DC side, such as the cable and distribution box, the capacitive discharge current is another abnormal condition that should be solved by RCD especially for the project with transformerless PV inverters.



Capacitive discharge current is associated with parasitic capacitance, which mainly come from two part.

The first, the solar module. Whom are familiar with the solar module will get the point that the sandwich structure is kinds of natural capacitor, i.e. the C_1, C_2, C_3 as below.



The Total parasitic capacitance is,

$$C_{total} = C_1 + C_2 + C_3$$

C_1, the parasitic capacitance due to the film of water on the glass, which is usually much larger than the others in wet environment. So

$$C_{total} \approx C_1$$

The C_{total} is proportional to the conductive surface (area of solar cell)present in your PV power plant.

The other, inverter.

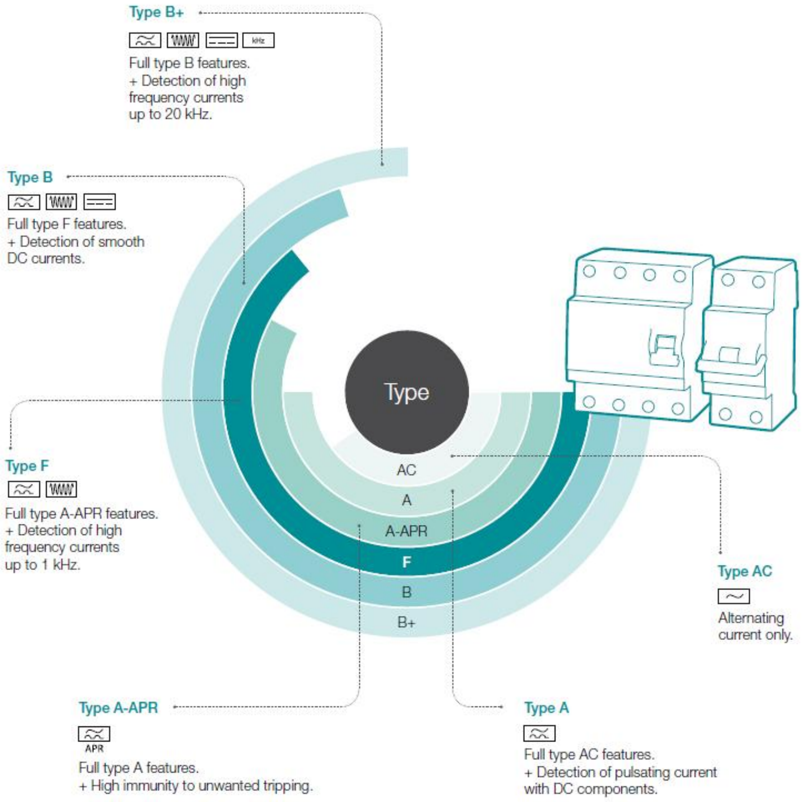
During operation, the DC bus is connected to the alternating current grid via the inverter. Thus, a portion of the alternating voltage amplitude arrives at the DC bus. The fluctuating voltage constantly changes the charge state of the parasitic PV capacitor (i.e. capacitance to PE). This is associated with a displacement current, which is proportional to the capacitance and the applied voltage amplitude.

Classification of RCD. And what do the type A, B, A(APR), B+ stand for?

According to the function included, RCDs can divide into:

Type	Explain
RCD	Generic term covering the devices incorporating residual current protection
RCCB	Residual current Circuit Breaker without overload or short circuit protection
RCBO	Residual current Circuit Breaker with overload and short circuit protection(MCB)

According to the wave form it could detect, i.e. type of residual current, RCD has Type AC, A, A(APR), F, B, B+ and so on. Please find more below.



There RCDs called Type APR in ABB portfolio is against unwanted tripping, which are called type G in Austrian.

Type B+ are mostly used to prevent fire protection risk as they are recommended by the association of German Insurance Companies.

RCD selecting recommendation for SolaX inverter

1. Our recommendations can't replace local associated codes.
2. Whether an RCD is necessary and which types may be used in general depends on the respective applicable installation regulations in each case.
3. Ask for service from qualified electrical technician once necessary.

1.1 Rated residual current, $I_{\Delta n}$

RCD sensitivity is expressed as the rated residual current, noted $I_{\Delta n}$.

Associated criteria,

No.	Criteria	Note
1.	High sensitivity: 6 – 10 – 30 mA Medium sensitivity: 0.1 – 0.3 – 0.5 – 1 A Low sensitivity: 3 –10 – 30 A RCDs for residential or similar applications are always high or medium sensitivity. High sensitivity is most often used for direct-contact protection (human protection, domestic installations), whereas medium sensitivity and in particular the 300 and 500 mA ratings are indispensable for fire protection. The other sensitivities are used for other needs such as protection against indirect contacts (mandatory in the TT system) or protection of machines.	Basic guidance.
2	If the installation is located in a barn or in wooden cabins, for example, DIN VDE 0100-482 (IEC 60364-4-42:2001-08) also applies. In that case, a residual-current device with a rated residual current of max. 300 mA is required for fire protection reasons.	Specifically for Germany.

SolaX's recommendation

Inverter Type	$I_{\Delta n}$, Minimum Value for one model

X1-Mini	300mA
X1-Boost	300mA
X3-Mic	300mA
X3-Pro (<20kW)	100mA
X3-Pro (≥20kW)	300mA
X3-Mega	300mA
X3-FORTH	300mA
X1-Hybrid	100mA
X3-Hybrid	300mA
X1-Fit	100mA
X3-Fit	100mA
X1-AC	100mA
X1-Vast	300mA
J1-ESS	100mA
A1-ESS	100mA
X3-ULTRA	300mA
X1-IES	300mA
X3-IES	300mA
X3-HYB-G4 PRO	300mA
X3-NEO-LV	300mA
ESS-AELIO	300mA
ESS-TRENE	300mA

1.2 Type A or Type B

All SolaX inverters have an integrated RCMU and relay measuring both AC and DC component. Due to the inverters' circuit design, they can't feed in direct residual current into the grid which comply with IEC60364-7-712:2012.

SolaX inverters are approved for use of Type A RCD.

Bibliography

1. Technical guide | 2015, Protection against earth faults with Residual Current Devices (ABB)